Finance Tracker

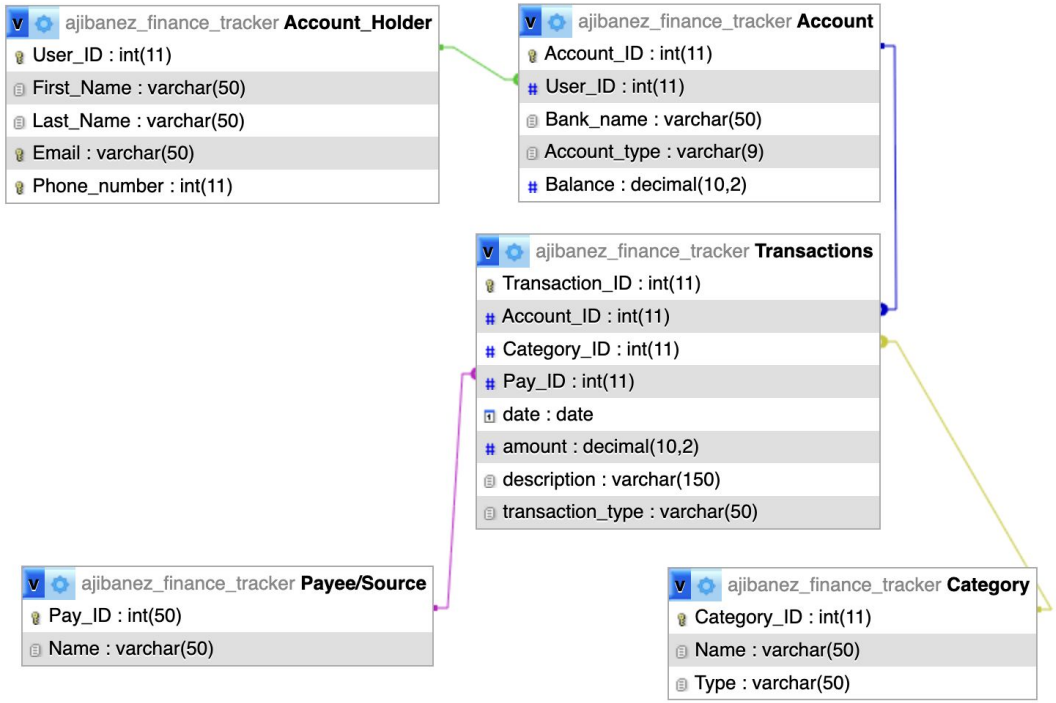
Group 5

Introduction

A database for users to track their finances

- Track income and expenses
- Manage multiple accounts
- Organize transactions by category
- Store and query financial data

Database Structure



Data Schema

Account Holder

user_id (PK)	first_name	last_name	email	phone_id
1	Benjamin	Jackson	benjamin.jackson1@example.com	326381088
2	Emma	Taylor	emma.taylor2@example.com	1713142197
3	Ava	Moore	ava.moore3@example.com	1626214580
4	Mia	Hernandez	mia.hernandez4@example.com	827498608
5	Benjamin	Moore	benjamin.moore5@example.com	961577524
6	Harper	Anderson	harper.anderson6@example.com	415554534
7	Ava	Thomas	ava.thomas7@example.com	974779629
8	Benjamin	Johnson	benjamin.johnson8@example.com	567795656
9	Noah	Martin	noah.martin9@example.com	1199700961
10	Harper	Rodriguez	harper.rodriguez10@example.com	2111737782
11	James	Martin	james.martin11@example.com	1156118306
12	Mia	Moore	mia.moore12@example.com	1316781548
13	Liam	Thomas	liam.thomas13@example.com	2106856929
14	Lucas	Moore	lucas.moore14@example.com	442921686
15	James	Miller	james.miller15@example.com	955916507
16	Liam	Johnson	liam.johnson16@example.com	610809873
17	Sophia	Davis	sophia.davis17@example.com	2032247696
18	Daniel	Johnson	daniel.johnson18@example.com	1049790610
19	Ethan	Johnson	ethan.johnson19@example.com	721338013
20	Amelia	Rodriguez	amelia.rodriguez20@example.com	477925308
21	Ethan	Davis	ethan.davis21@example.com	1569472235
22	Amelia	Hernandez	amelia.hernandez22@example.com	1400503513
23	Emma	Jackson	emma.jackson23@example.com	303315752
24	Ethan	Moore	ethan.moore24@example.com	819971385
25	Benjamin	Johnson	benjamin.johnson25@example.com	1216317090
26	Mia	Williams	mia.williams26@example.com	565697089
27	Ethan	Jackson	ethan.jackson27@example.com	1724239752
28	Lucas	Williams	lucas.williams28@example.com	1338290561
29	Sophia	Martin	sophia.martin29@example.com	1624327337

Account

account_id (PK)	user_id (FK→Account_Holder)	bank_name	account_type	balance
1	1	U.S. Bank	Checking	$1,186.97
2	2	Chase	Savings	$1,440.78
3	3	Capital One	Money Market	$2,172.70
4	4	Truist	Savings	$516.01
5	5	TD Bank	Checking	$3,700.98
6	6	Bank of America	Money Market	$2,043.64
7	7	Wells Fargo	Checking	$4,494.09
8	8	U.S. Bank	Money Market	$2,919.90
9	9	Wells Fargo	Savings	$1,610.19
10	10	U.S. Bank	Money Market	$1,441.39
11	11	Truist	Savings	$2,026.33
12	12	Capital One	Savings	$835.42
13	13	Citibank	Checking	$1,715.38
14	14	Wells Fargo	Money Market	$3,789.49
15	15	Truist	Savings	$3,803.63
16	16	Ally	Money Market	$1,582.86
17	17	Truist	Savings	$2,111.07
18	18	Ally	Checking	$537.49
19	19	Ally	Money Market	$3,587.41
20	20	Ally	Money Market	$542.48
21	21	Capital One	Savings	$3,242.54
22	22	Citibank	Savings	$1,988.23
23	23	Wells Fargo	Checking	$3,819.78
24	24	Chase	Savings	$4,026.89
25	25	Ally	Savings	$2,962.92
26	26	Truist	Money Market	$3,776.18
27	27	U.S. Bank	Money Market	$3,159.92
28	28	Ally	Money Market	$580.94
29	29	Ally	Savings	$3,056.40

Transaction

	Transaction_ID	Account_ID	Category_ID	Pay_ID	date	amount	description	transaction_type
Edit Copy Delete	1	50	4	7	2026-01-11	1536.00	Dropshipping	Deposit
Edit Copy Delete	2	96	6	7	2025-12-02	1263.00	Misc	Withdrawal
Edit Copy Delete	3	82	4	7	2026-01-16	4888.00	Dropshipping	Deposit
Edit Copy Delete	4	38	2	5	2025-12-31	110.00	Groceries	Withdrawal
Edit Copy Delete	5	45	7	3	2025-12-03	377.00	Utilities	Withdrawal
Edit Copy Delete	6	22	2	5	2026-01-06	225.00	Groceries	Withdrawal
Edit Copy Delete	7	26	3	4	2026-01-21	1809.00	Rent	Withdrawal
Edit Copy Delete	8	29	3	4	2025-09-02	1574.00	Rent	Withdrawal
Edit Copy Delete	9	85	2	2	2025-09-07	222.00	Groceries	Withdrawal
Edit Copy Delete	10	94	2	5	2025-12-01	602.00	Groceries	Withdrawal
Edit Copy Delete	11	13	6	8	2026-01-04	2949.00	Misc	Withdrawal
Edit Copy Delete	12	68	6	5	2025-09-15	3656.00	Misc	Withdrawal
Edit Copy Delete	13	14	1	1	2025-10-02	267.00	Salary	Deposit
Edit Copy Delete	14	47	2	5	2025-10-31	580.00	Groceries	Withdrawal
Edit Copy Delete	15	99	1	1	2025-10-04	994.00	Salary	Deposit
Edit Copy Delete	16	93	7	3	2025-09-01	127.00	Utilities	Withdrawal
Edit Copy Delete	17	66	6	6	2025-11-19	2375.00	Misc	Withdrawal
Edit Copy Delete	18	61	6	2	2025-09-14	2793.00	Misc	Withdrawal
Edit Copy Delete	19	83	6	2	2025-10-25	863.00	Misc	Withdrawal
Edit Copy Delete	20	62	6	8	2025-12-03	4865.00	Misc	Withdrawal
Edit Copy Delete	21	60	2	6	2026-01-12	60.00	Groceries	Withdrawal
Edit Copy Delete	22	16	6	2	2025-09-20	1646.00	Misc	Withdrawal
Console it Copy Delete	23	84	1	1	2025-12-18	3799.00	Salary	Deposit

Payee/Sources

payee_id (PK)	payee_name
1	Dropshipping Income
2	Employer Direct Deposit
3	Grocery Stores
4	Landlord/Property Management
5	Miscellaneous Vendors
6	Utility Companies
7	Gaming Platforms
8	Educational Institutions
9	Insurance Companies
10	Transportation Services

Category

category_id (PK)	category_name	category_type
1	Salary	Income
2	Groceries	Expense
3	Rent	Expense
4	Dropshipping	Income
5	Games	Expense
6	Miscellanious	Expense
7	Utilities	Expense
8	Transportation	Expense
9	Education	Expense
10	Insurance	Expense

Database Queries

Execute a live DDL query (i.e., **CREATE**, **DROP**, **ALTER**) on your database.

```
ALTER TABLE
Account_Holder
MODIFY Phone_number
VARCHAR(15);
```

Execute a live DML query (i.e., **INSERT**, **UPDATE**, **DELETE**) on your database.

```
UPDATE Account_Holder
SET Phone_number = 9999999999
WHERE First_Name = 'Emma'
AND Last_Name = 'Taylor';
```

Database Queries Ⅱ

```
SELECT * FROM
Transactions WHERE
Category_ID = 1
```

```
SELECT * FROM
`Transactions` WHERE
Category_ID = 7
```

```
SELECT description, amount,
ROUND(amount*0.10,2)AS Tax,
ROUND(amount-(amount*0.10),2)AS
Amount_Before_Tax
FROM Transactions;
```

```
SELECT Last_Name, COUNT(*)
AS NumAccounts

FROM Account_Holder

WHERE Last_Name IN
("Johnson","Garcia")

GROUP BY Last_Name;
```

```
SELECT

Account_Holder.First_Name,
Account_Holder.Last_Name,Category.Name,Transactions.
Amount,Transactions.Transaction_Type

FROM Transactions

JOIN Account ON Transactions.Account_ID =
Account.Account_ID

JOIN Account_Holder Account_Holder ON
Account.User_ID = Account_Holder.User_ID

JOIN Category ON Transactions.Category_ID =
Category.Category_ID

WHERE Account_Holder.Last_Name IN
('Johnson','Garcia')

ORDER BY Account_Holder.Last_Name,
Transactions.Amount DESC;
```